



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarian with username and password"
- "Validate username uniqueness"
- "Enforce strong password policy"
- "Authenticate librarian credentials"
- "Issue and validate sessions"
- "Handle credentials securely"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books by title"
- "Prevent duplicate book registration"
- "Delete registered books"
- "List all registered books"
- "Consult book by identifier"
- "Consult book by title"
- "List books available for loan"
- "Validate book existence and availability"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans"
- "Prevent duplicate reader registration"
- "Delete registered readers"
- "List all registered readers"
- "Consult reader by identifier"
- "Consult reader by name"
- "Validate reader existence"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Service"

responsibilities:

- "Register book loans for registered readers"
- "Record loan date"
- "Prevent loans for unregistered readers"
- "Prevent loans for unregistered books"
- "Prevent loans for already borrowed books"
- "Allow multiple loans per reader"
- "Ensure one book loaned to one reader at a time"
- "Validate reader and book data during loan registration"
- "Remove loan links when book or reader is deleted"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/loan-service"

- name: "Environment Management Service"

responsibilities:

- "Clear data and base structure"
- "Recreate base structure"
- "Restore environment to clean state"
- "Coordinate reset operations across functional areas"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Loan Service"
- "Authentication Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum two integers"
- "Calculate Fibonacci sequence using multiple approaches"
- "Calculate Ackermann function using multiple approaches"
- "Provide service catalog for endpoint testing"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Record main operations from business capabilities"
- "Support traceability and problem analysis"
- "Maintain decoupled event recording mechanism"

dependencies:

- ""

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"

to: "Audit Service"

- from: "Book Catalog Service"

to: "Audit Service"

- from: "Reader Management Service"

to: "Audit Service"

- from: "Loan Service"

to: "Book Catalog Service"

- from: "Loan Service"

to: "Reader Management Service"

- from: "Loan Service"

to: "Audit Service"

- from: "Environment Management Service"

to: "Book Catalog Service"

- from: "Environment Management Service"

to: "Reader Management Service"

- from: "Environment Management Service"

to: "Loan Service"

- from: "Environment Management Service"

to: "Authentication Service"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Validate customer login credentials"
- "Create usage session after successful authentication"
- "Validate active session for restricted access"
- "End customer session through logout"
- "Invalidate session immediately on logout"
- "Manage session expiration based on configured validity period"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Search flights by origin airport destination airport and travel date"
- "Support one-way and round-trip searches"
- "Present flight options for each trip segment"
- "Show departure time arrival time and reference fare values for each flight"
- "Manage registered airports"
- "Maintain routes schedules and reference fares"

dependencies:

- ""

suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create booking for one-way flight"
- "Create booking for outbound and return flights in same operation"
- "Generate booking identifier"
- "Record booking date"
- "Allow customers to consult existing bookings"
- "Present list of completed bookings"
- "Cancel selected booking"
- "Remove cancelled booking from active customer listing"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Allow customers to consult their own profile"
- "Show customer contact data including phone phone type and address"
- "Update contact and address data"
- "Require current password to confirm profile changes"
- "Prevent customers from viewing or changing another customer profile"
- "Maintain customer registration and contact data"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Show customer loyalty program status"
- "Show number of miles accumulated in the year"
- "Show total accumulated miles"
- "Provide read-only loyalty information"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Allow initial loading or reloading of application database"
- "Define number of sample customers to be generated"
- "Recreate main operational data including flights customers sessions and test bookings"
- "Provide summarized operational information such as numbers of customers sessions flights segments booki
- "Prepare and verify database for tests demonstrations or operational reset"
- "Provide basic operational indicators"

dependencies:

- "Authentication Service"
- "Flight Search Service"
- "Booking Service"
- "Customer Profile Service"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"
to: "Authentication Service"
- from: "Booking Service"
to: "Flight Search Service"
- from: "Customer Profile Service"
to: "Authentication Service"
- from: "Loyalty Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Flight Search Service"
- from: "Environment Administration Service"
to: "Booking Service"
- from: "Environment Administration Service"
to: "Customer Profile Service"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register new cargo with origin destination and deadline"
- "Assign unique tracking identifier"
- "Validate origin and destination are different and registered locations"
- "Maintain fixed cargo origin"
- "Allow cargo registration without defined route"
- "Change cargo destination"
- "Validate new destination differs from origin and current destination"
- "Trigger route reevaluation when destination changes"
- "Provide cargo information for consultation"

dependencies:

- "Location Service"
- "Route Planning Service"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Location Service"

responsibilities:

- "Maintain registry of logistics locations"
- "Validate location existence for cargo registration"
- "Provide available locations for origin and destination selection"

dependencies:

- ""

suggested_endpoints:

- "/api/location-service"

- name: "Route Planning Service"

responsibilities:

- "Retrieve candidate routes compatible with cargo needs"
- "Validate routes respect origin destination and arrival deadline"
- "Present route options to user"
- "Assign selected route to cargo"
- "Display route legs with load points unload points voyages and dates"
- "Mark cargo as not routed when no route assigned"
- "Inform when no feasible route exists"
- "Reevaluate route when cargo destination changes"
- "Mark cargo for new routing when current route invalid"
- "Replan routes for off-route or changed cargos"

dependencies:

- "Voyage Service"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Voyage Service"

responsibilities:

- "Maintain voyage information"
- "Validate voyage existence for load and unload events"
- "Provide voyage details for route planning"

dependencies:

- ""

suggested_endpoints:

- "/api/voyage-service"

- name: "Event Processing Service"

responsibilities:

- "Register logistics events of types reception load unload customs and pickup"
- "Associate events with existing cargo"
- "Record event location and occurrence date"
- "Associate load and unload events with corresponding voyage"
- "Reject events with nonexistent cargo location or voyage"
- "Validate event data before processing"
- "Update cargo operational situation after valid event"
- "Recalculate transport status last known location next expected step and plan adherence"
- "Mark cargo as off route when event differs from planned route"
- "Mark cargo as arrived when unloaded at final destination"
- "Mark cargo as complete when pickup registered"
- "Maintain event occurrence and registration timestamps"
- "Identify rejected registration attempts for analysis"

dependencies:

- "Location Service"
- "Voyage Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/event-processing-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consolidate cargo operational status and history"
- "Provide cargo consultation by tracking identifier"
- "Show current cargo situation"
- "Inform last known location"
- "Calculate and inform estimated arrival for cargos with valid route"
- "Maintain and display logistics event history"
- "Distinguish expected from unexpected events"
- "Signal when cargo is off expected route"
- "Provide geographic view with origin destination and last location"
- "Reconstruct cargo logistics cycle from history"
- "Distinguish delivered in-transit and deviated cargos"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Event Processing Service"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Receive logistics events through operational interface"
- "Receive logistics events through integration service channel"
- "Process logistics event input files periodically"
- "Provide consolidated cargo view for monitoring channels"
- "Provide standardized cargo information across channels"
- "Exchange cargo information in standardized formats"
- "Maintain consistent tracking model across all channels"

dependencies:

- "Event Processing Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Cargo Management Service"
to: "Location Service"
- from: "Cargo Management Service"
to: "Route Planning Service"
- from: "Route Planning Service"
to: "Voyage Service"
- from: "Event Processing Service"
to: "Location Service"
- from: "Event Processing Service"
to: "Voyage Service"
- from: "Event Processing Service"
to: "Cargo Tracking Service"
- from: "Cargo Tracking Service"
to: "Cargo Management Service"
- from: "Cargo Tracking Service"
to: "Route Planning Service"
- from: "Cargo Tracking Service"
to: "Event Processing Service"
- from: "Integration Service"
to: "Event Processing Service"
- from: "Integration Service"
to: "Cargo Tracking Service"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investor credentials"
- "Start investor session"
- "End investor session through logout"
- "Validate investor authentication status"

dependencies:

- "Investor Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Investor Account Service"

responsibilities:

- "Register new investors"
- "Validate unique investor identification"
- "Maintain investor account data including balance and access statistics"
- "Consult investor registration and financial data"
- "Update investor profile"
- "Validate profile update fields"
- "Validate password confirmation match"

dependencies:

- "Authentication Service"
- "Trading Service"
- "Portfolio Service"

suggested_endpoints:

- "/api/investor-account-service"

- name: "Market Data Service"

responsibilities:

- "Query asset quotes by code"
- "Query multiple assets simultaneously"
- "Provide asset information including company name and price data"
- "Provide consolidated market summary with performance index and top assets"
- "Handle invalid asset codes"
- "Update traded volume and market indicators after operations"

dependencies:

- "Trading Service"

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create buy orders for assets"
- "Create sell orders for portfolio positions"
- "Generate individual trading orders with identifier and metadata"
- "Apply fixed operation fees"
- "Debit investor balance for buy operations"
- "Credit investor balance for sell operations"
- "Process orders immediately or later"
- "Track order status through lifecycle"
- "Prevent duplicate order completion"
- "Cancel sales for unavailable positions"
- "Confirm order submission"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Portfolio Service"
- "Order History Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor portfolio positions"
- "Register new positions from completed buy orders"
- "Remove positions from completed sell orders"
- "Consult investor portfolio"
- "Calculate position metrics including gain or loss"
- "Calculate total invested amount and market value"
- "Show consolidated financial position summary"
- "Initiate sell orders from portfolio positions"

dependencies:

- "Trading Service"
- "Investor Account Service"
- "Market Data Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Order History Service"

responsibilities:

- "Maintain history of investor orders"
- "Show summarized view of recent orders"
- "Provide expanded order history consultation"
- "Alert investor about completed unviewed orders"

dependencies:

- "Trading Service"

suggested_endpoints:

- "/api/order-history-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters including processing mode and limits"
- "Enable or disable completed order alerts"
- "Prepare or rebuild simulation database"
- "Reset simulation operational data"
- "Provide execution statistics"

dependencies:

- "Trading Service"
- "Order History Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Investor Account Service"
- from: "Investor Account Service"
to: "Authentication Service"
- from: "Investor Account Service"
to: "Trading Service"
- from: "Investor Account Service"
to: "Portfolio Service"
- from: "Market Data Service"
to: "Trading Service"
- from: "Trading Service"
to: "Investor Account Service"
- from: "Trading Service"
to: "Market Data Service"
- from: "Trading Service"
to: "Portfolio Service"
- from: "Trading Service"
to: "Order History Service"
- from: "Portfolio Service"
to: "Trading Service"
- from: "Portfolio Service"
to: "Investor Account Service"
- from: "Portfolio Service"
to: "Market Data Service"
- from: "Order History Service"
to: "Trading Service"
- from: "Simulation Administration Service"
to: "Trading Service"
- from: "Simulation Administration Service"

to: "Order History Service"

SYSTEM: Jokul

system: "Jokul"

services:

interactions:

SYSTEM: JpetStore

system: "JpetStore"

services:

interactions:

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

interactions:

SYSTEM: TNTConcept

system: "TNTConcept"

services:

interactions: